utility companies. This is a cheaper option compared to a battery storage system that is generally used for charging overnight.

The cost of setting up a solar-powered charging station depends on factors like installation cost, hardware equipment, cost of electricity and maintenance in the long run. The cost of purchasing a solar array is based on the manufacturer and selection of solar panels. For those who already have a solar array installed at their homes, they just need to have an EV charging station.

Besides better air quality and reduced dependence on fossil fuels, low costs are incurred and there is a reduced tendency of health-related risks. The load on conventional grids also gets reduced. Apart from this, a large-scale implementation will increase employment owing to the need for installation, maintenance, and operation.

Considering the benefits and viability of such a system, many businesses are investing in this concept. For instance, Tesla Motors, a subsidiary of Tesla is constructing solar-powered charging stations in convenient locations for its EV customers.

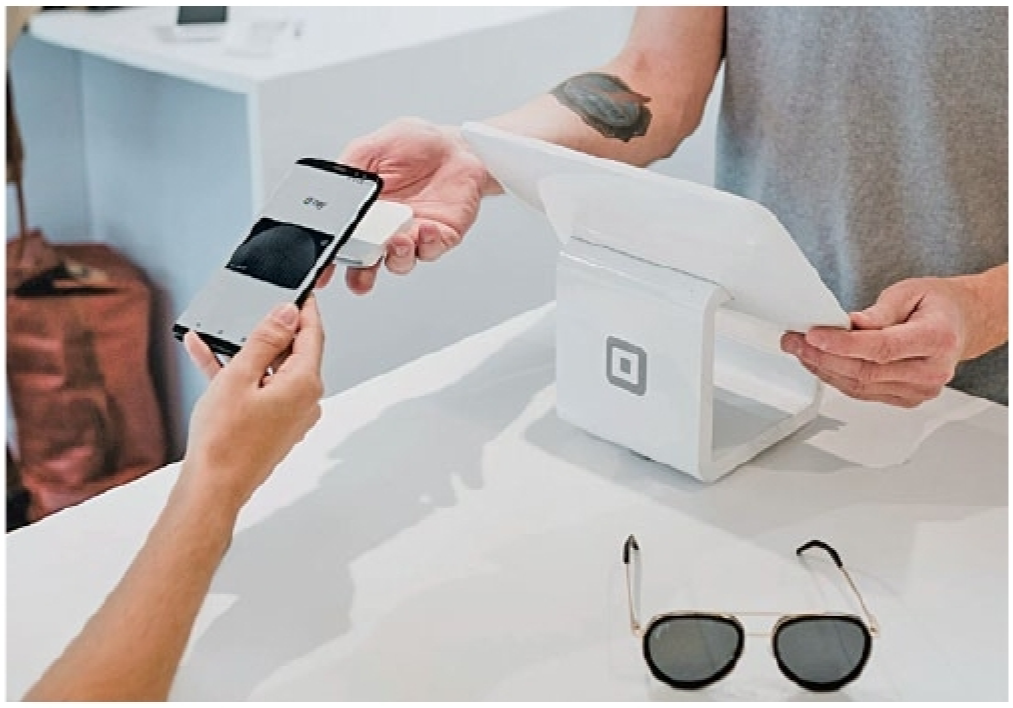
Although the payback period is long, collaborations and government policies can decrease it and help in establishing a robust charging infrastructure. In India, the government has de-licensed public charging stations business for EVs. In 2019, BHEL announced that it will be setting up five solar-based EV charging stations at Haryana Tourism Corporation's resorts on the Delhi-Chandigarh highway. The organisation also developed a central monitoring system and a mobile app for EV charging. Japanese electronics company Panasonic has also been working to set up nearly one lakh charging stations for EVs across twenty-five top Indian cities.

*—Ayushee Sharma*

# 2 SCOPE AND IMPACT OF CLOUD COMPUTING ON BANKING

**With the introduction of cloud computing, things have become easier and much accessible in banking and financial sectors. Cloud computing is a hybrid technology that helps in storing and recovering huge data of the banks, besides cost reduction and better customer services**

Cloud computing is a hybrid technology of computing various services like software, servers, storage, databases, networking, analytics and many more over the Internet. It can be easily deployed and integrated with all the services of the banking system, which decreases the time and effort of the bankers and customers. It has made many things easier in banking and finance sectors like interoperability, 24 × 7 access, secure storage, etc. Using software or banking apps from any device from any location has become seamless.



On-demand cloud services are delivered with a pay-as-you-go billing model that leads to significant cost reduction. Cloud computing helps in storing, backup and recovering huge data of the banks. It has enabled the banks to focus more on the customer-centric model and digitalising trading and wealth. Every information stored in the cloud is safe with end-to-end protection.

Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM) are the most popular software in cloud computing. These software allow the banks and financial institutions to secure the data and provide better support to the customers. Some cloud providers who provide hybrid cloud computing servers to the companies are Amazon Web Services, Google Virtual Cloud, Microsoft Azure, Aliyun, and IBM Bluemix.

Banks can access the cloud any time as required, which means they can utilise it more flexibly. This increases the banks' ability to relieve their staff from the administration of IT infrastructure and reallocate resources for more innovative banking solutions. During peak customer demand hours, the cloud can allow banks to scale computing resources according to the requirements of the consumers. The scalability of the cloud means it can scan potentially thousands of transactions per second to improve security against fraud and money laundering.

In a bid to ensure secure transactions and smooth customer experience, banks need 100 per cent uptime.